1. INTRODUCTION

1.1 Background and Aims

Separations technology is important in almost every chemical engineering process. This unit builds on material taught in CET I on equilibrium staged processes to introduce additional separation processes. The course provides fundamental knowledge to calculate the performance of continuous contacting separation processes in which mass transfer rates are important.

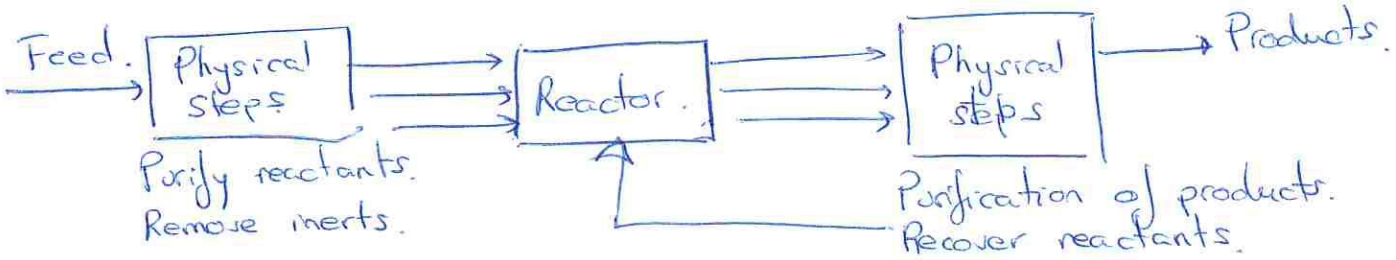
Aims:

On completing this course and the associated problem sheets, students should be able to:

- Understand the principles of separation processes where performance is dictated by the rate mass transfer rather than equilibrium stages.
- Understand how mass transfer affects the performance of equilibrium gas absorption processes and how it needs to be considered for the design of absorption columns.
- Understand the principles of drying processes and perform calculations for the design of drying equipment.
- Understand the principles of membrane separation processes and perform calculations on the rates of flux using underlying principles.
- Understand the principles underlying adsorption processes in gas and liquid systems and perform calculations of the performance of adsorption systems.

1.2 Separation Philosophy

A typical chemical plant normally consists of a reactor, which will determine the sequence of unit operations upstream (e.g. to purify reactants, remove inert/poison compounds, etc.) and downstream (e.g. recover unreacted feed, purification of products, etc.):



Separation: The selective movement of the components in a feedstock to separate locations.

There are five basic separation methods:

i. Separation by *phase creation* (Δ volatilities or Δ boiling points)
Distillation or crystallization.

ii. Separation by *phase addition* (Δ solubilities)
L-L extraction - absorption.

iii. Separation by *barrier* (Δ permeabilities)
Membranes.

iv. Separation by *solid agent* (Δ adsorbabilities)
Adsorption.

v. Separation by *force field* or *gradient* (Δ electrical charges, diffusivities)
Centrifugal forces, electrical fields.

Each of these separation methods exploit differences in:

i. Molecular properties: e.g. molecular weight, dipole moment, electrical charge, hydrophobicity. etc

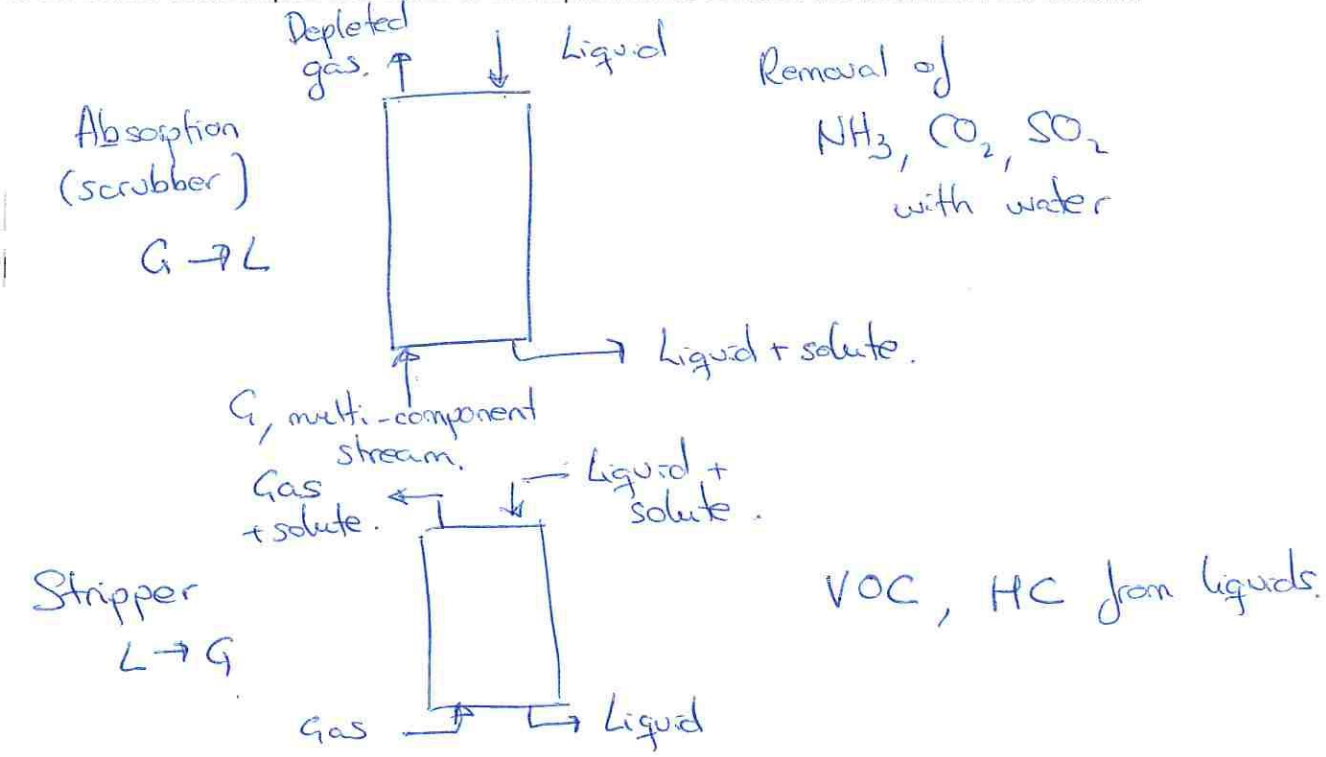
ii. Thermodynamic and transport properties: e.g. vapour pressure, solubility, volatility, diffusivity, adsorptivity

These properties are usually tabulated in handbooks, journal articles and reference books.

2. GAS ABSORPTION

Absorption is the removal of a solute in an incoming gas to a liquid. The mol fraction of the solute in the gas decreases as the gas moves towards the top of the column. The mol fraction of the solute in the liquid increases as the liquid moves towards the bottom of the column.

Stripping is the removal of solute from an incoming liquid to a gas. The mol fraction of the solute in the gas increases as the gas moves towards the top of the column. The mol fraction of the solute in the liquid decreases as the liquid moves towards the bottom of the column.

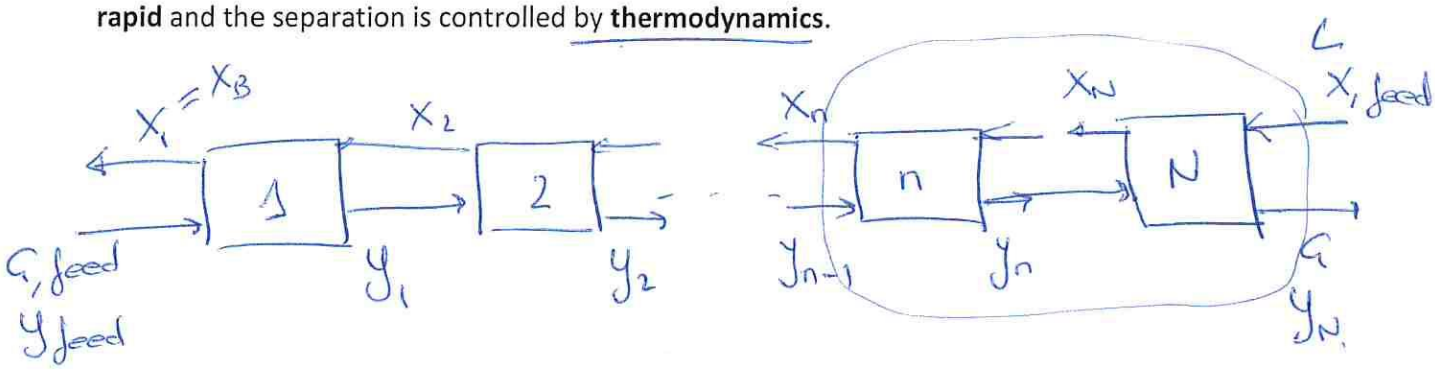


In both cases, absorption and stripping can be treated as:

- Physical process: when chemical reactions have no appreciable effects. For example, absorption of acetone/air or ammonia/air in water.
- Chemical processes: when a chemical reaction influences the actual rate of absorption. For example, nitrogen oxides to absorb in water to form nitric acid, or carbon dioxide absorbing in sodium hydroxide.

When two phases are brought into contact, the transfer of one or more components from one phase into another is based on interphase mass transfer controlled largely by rates of diffusion. If the contact takes place during enough time, they eventually reach equilibrium.

Last year, in CEBT 1B, you studied the case where separation is carried out in equipment involving exchange between **discrete stages.** In these cases, the partition of components is **rapid** and the separation is controlled by **thermodynamics.**



These discrete stages may be plates in distillation absorption columns, or mixed settler units.

To calculate the number of stages needed to achieve a certain level of separation, you learnt the McCabe-Thiele construction of counter-current cascades of equilibrium stages in absorption systems.

The **operating line** is given by a material balance on the control volume (assuming that the system is dilute, so G and L were constant from stage to stage).

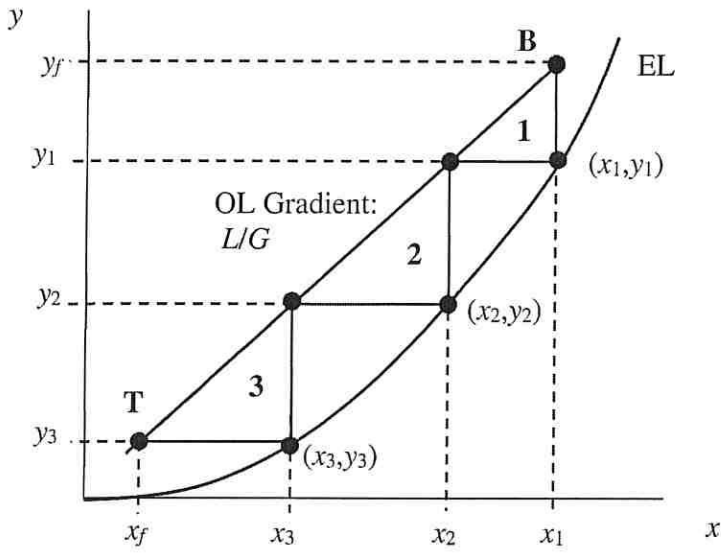
$$y_{n-1} = \frac{L}{G} x_n + \left(y_N - \frac{L}{G} x_f \right) \qquad (2.1)$$

The **equilibrium line**, given for example by Henry's law:

$$y = k \cdot x \qquad (2.2)$$

This *operating line* (OL) has slope *L/G* and connects all the points (x_n, y_{n-1}) to the point (x_f, y_N) and so relates the compositions of passing streams.

It is possible to determine the number of stages, *N*, required to affect the separation using the *McCabe-Thiele construction*, as follows (*design calculation*):



However, **continuous contact** between the phases (rather than stage-wise) is also popular. In these cases, **equilibrium between phases does not occur**, and design is based upon the **rate of mass transfer** between the phases.

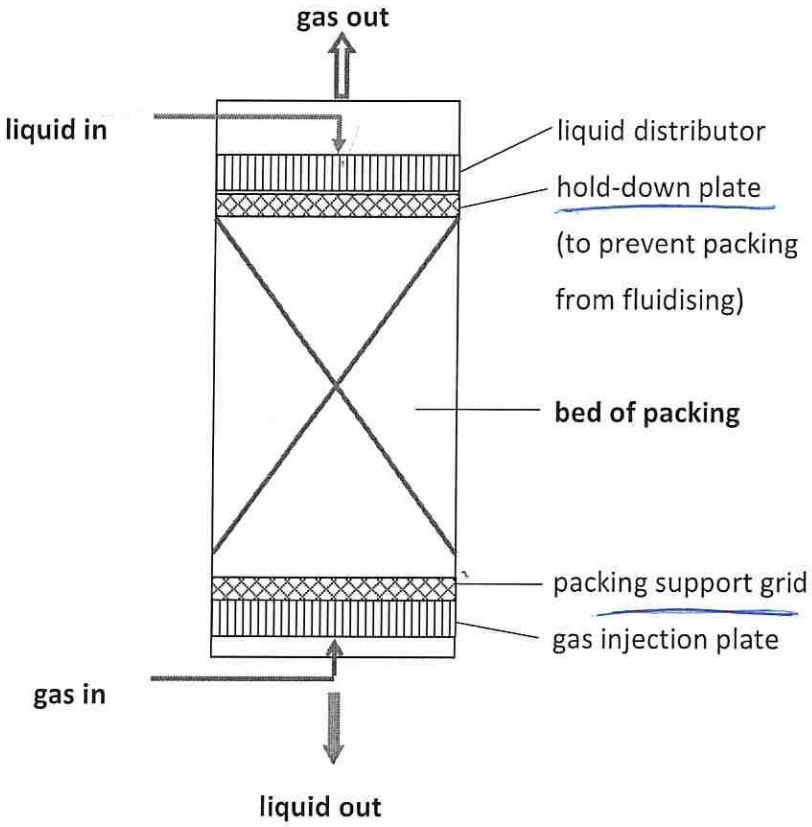
If resistances to mass transfer are significant, then **kinetics** determine the amount transferred.

This course deals with **rate-limited** processes under ideal conditions of **plug flow**, **no dispersion**, negligible thermal effects (**isothermal**) and **dilute systems** (*i.e.* a small amount of soluble component is present in the gas or liquid flow, so that the change in molar flow rates of gas and liquid between inlet and outlet is negligible).

2.1. SEPARATION EQUIPMENT

There are two main types of equipment for gas absorption processes: packed columns and plate columns.

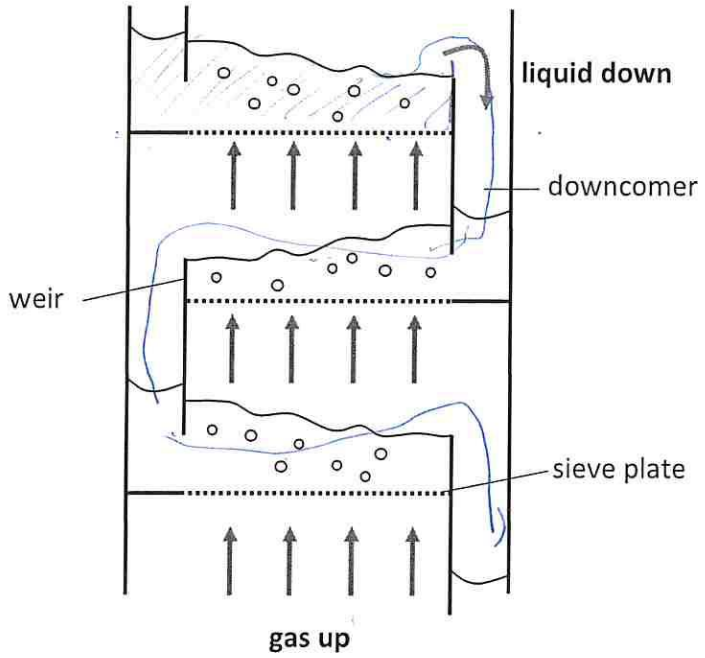
a. **Packed columns:** are used for both gas absorption and desorption. The packing material provides a large interfacial area for mass transfer and can consist of **random** (e.g. rings, saddles) or **structured** (e.g. grid packing – used mainly in water cooling towers). Structured packings offer greater transfer area per unit volume and better flow patterns, although at a higher cost per unit volume.



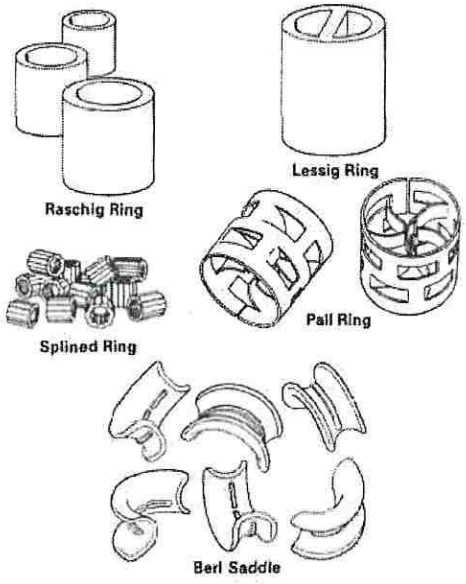
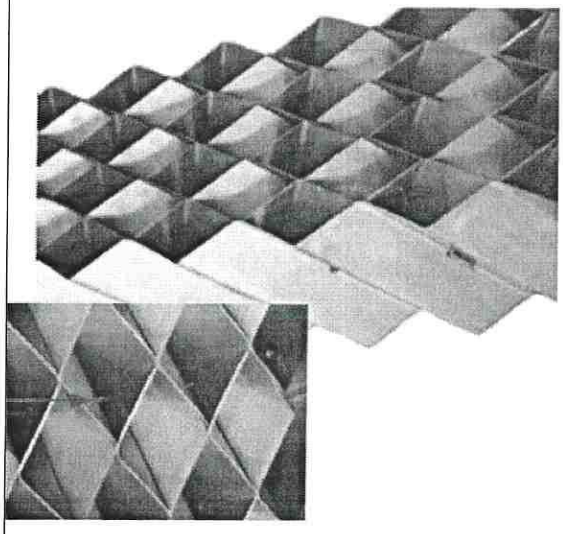
Even distribution of both gas and liquid over the cross-sectional area of the bed is critical for good performance; channelling of one or both phases can be a major problem, especially with random packings. Large columns may thus feature **liquid redistribution sections** (packing support grid + liquid distributor).

Industrial columns may range up to **5 m in diameter and 30 m in height**.

b. Plate columns: use sieve trays, bubble cap trays or other tray designs, as for distillation



Types of packing:

Random packing	Corrugated grid (structured packing)
 Figure 6-8. Various types of packing. (http://www.oilngasseparator.info)	 (http://www.amistco.com)

Comparison of plate vs packed columns:

	Packed	**Plate**
Flow rates	Liquid hold-up is smaller (≈ 25% of the packing mass) – useful for minimising inventory of flammable or toxic liquids	Can handle wider ranges, especially low liquid flow rates
Design	Particularly useful for small diameter columns (< 0.6 m) since less expensive (plates are difficult to install in small columns) Packing can be replaced rather than cleaned *in situ* Cheaper to construct when handling corrosive liquids and vapours	Number of plates can be established with more certainty than packing height Withdrawal of side streams is easier Normally shorter
Heat transfer		Cooling and heating are easier (coils can be inserted in the plates)
Operation	Pressure drops are lower, therefore suitable for vacuum columns Good at handling foaming systems	Easier to clean

Wetted wall columns: They are purely lab devices which consist of a thin film of liquid which runs down the walls of a column whilst gas moves vertically upwards through the centre. Any soluble component in the gas stream will progressively dissolve in the falling liquid film as the gas moves up. It is used to study gas absorption since the surface area of the gas-liquid interface is clearly defined but small, and so wetted wall columns are not of interest industrially.

Used to calculate mass transfer rates



2.2 EQUILIBRIUM RELATIONSHIPS

Equilibrium relationships describe the partition of a component in a two-phase system.

There exist many ways of describing that relationship:

$C_g = H_c \cdot C_L$

C_g: gas phase.

H_c: dimensionless

C_L: concentration of species in liquid phase (bulk).

For dilute systems, we assume that $H_c \neq f(C_g, C_l)$, and that H_c is a type of **Henry's Law constant.**

Although H_c is dimensionless, it is different from the other dimensionless parameter K in an alternative description of equilibrium, namely:

$y = K \cdot x$

$K \rightarrow H_m / K_m$

$p = H' \cdot C_L$ $\quad H' = m/kmol$

$p = H'' \cdot x$ $\quad H'' = 1/m^2$

Some useful conversions:

$$x \equiv \frac{C_l}{\rho_{ml}}$$ ρ_{ml} = **molar density of liquid phase [kmol m^{-3}] = ρ_l/RMM**

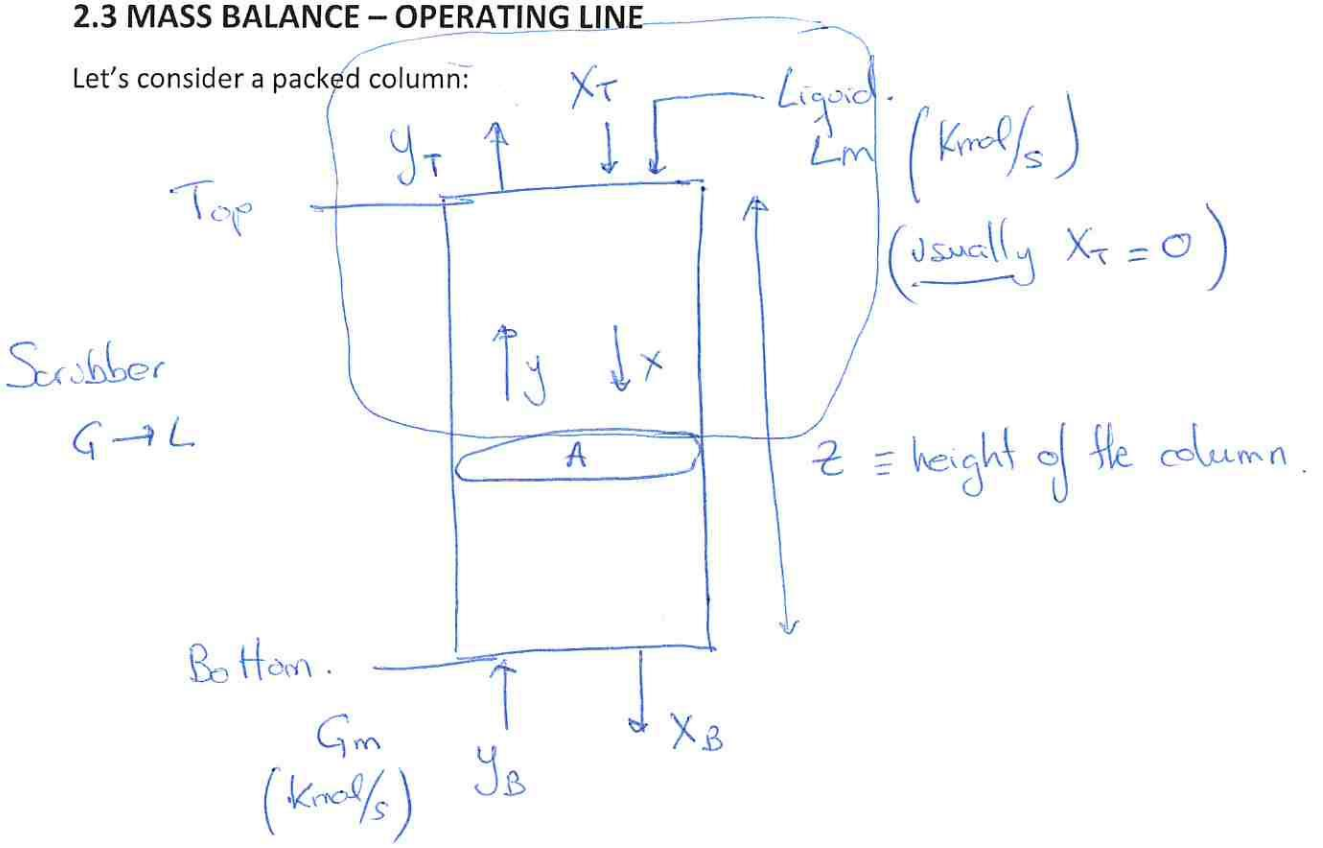
$$y \equiv \frac{p}{P_T} = \frac{C_g}{\rho_{mg}}$$ ρ_{mg} = **molar density of gas phase (ALL SPECIES)**

$= P_T/R_G T$ for ideal gas, *i.e.* **total gas conc.**

So if $H_c = \frac{C_g}{C_l}$ and $H_m = \frac{y}{x}$, then: $\boxed{H_c = \left(\frac{\rho_{mg}}{\rho_{ml}}\right) H_m}$

2.3 MASS BALANCE – OPERATING LINE

Let's consider a packed column:



For a dilute system, L_m and G_m are approximately constant over the column.

Molar balance [kmol s^{-1}] on component over **entire column**:

$$IN = OUT$$

$$G_m \cdot y_B + L_m \cdot X_T = G_m \cdot y_T + L_m \cdot X_B$$

$$\underbrace{G_m (y_B - y_T)}_{\text{Kmol transferred from gas}} = \underbrace{L_m (X_B - X_T)}_{\text{Kmol received by liquid}} = \underbrace{Q_m}_{\text{Overall rate of transfer of component i}} \left(\text{Kmol/s}\right)$$

$Q_m \ll L_m$ and G_m

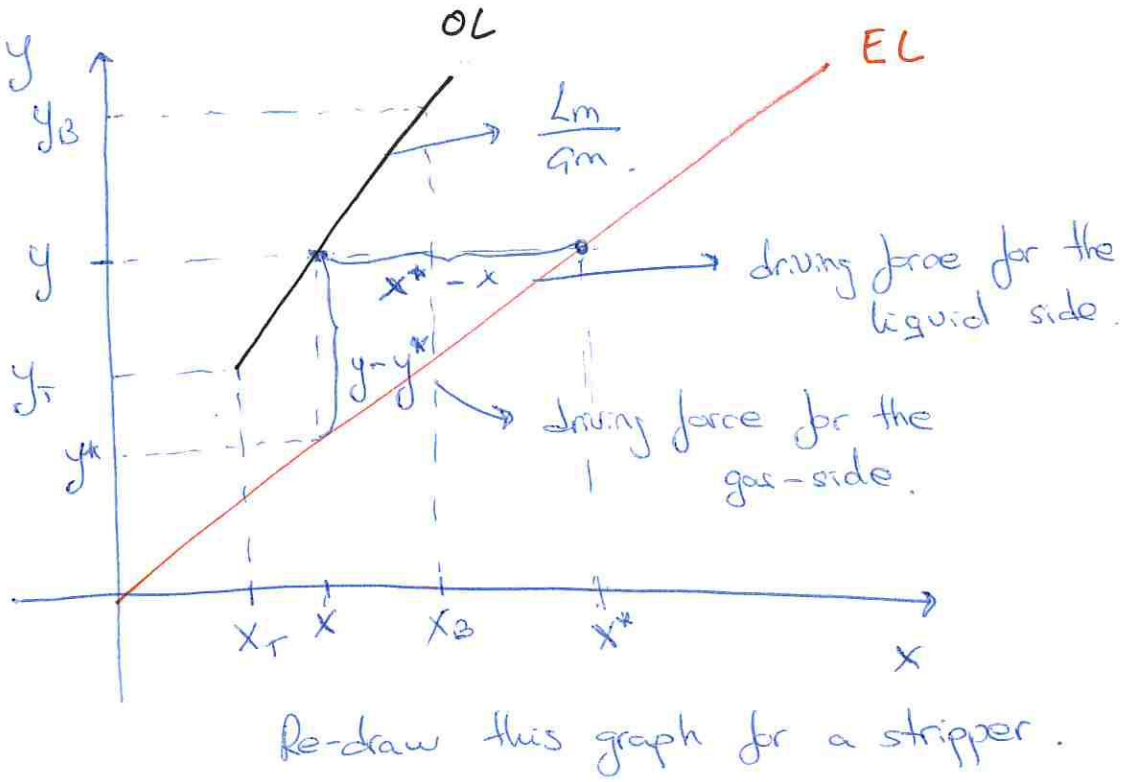
$$G_m (y - y_T) = L_m (x - x_T)$$

Solve for y: $$y = \left(\frac{L_m}{G_m}\right) \cdot x + \left(y_T - \frac{L_m}{G_m} \cdot x_T\right)$$

OPERATING LINE

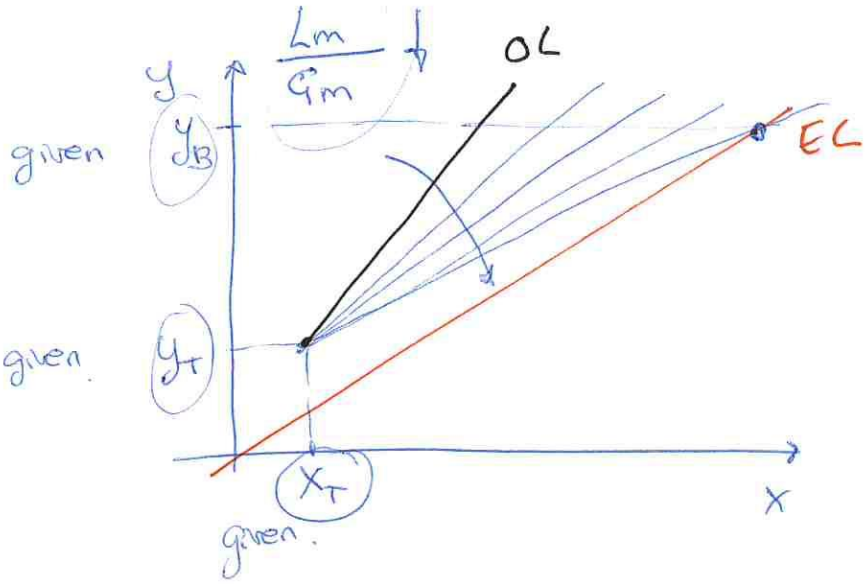
cf. equation of a straight line: **y = (gradient) x + constant**

A typical case for an absorber (scrubber) (gas being depleted, $y_B > y_T$):



2.3.1. DESIGN FLOW RATES AND THE PINCH

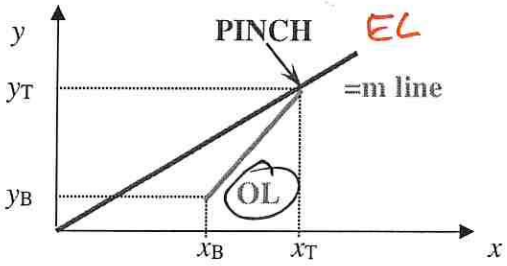
One flow rate is usually known, *e.g.* it is necessary to scrub a known gas stream of an impurity. The design problem is then to determine the **flow rate of the other stream**. Consider an operating line for an absorber whose gradient (L_m/G_m) decreases (so if G_m is fixed, L_m is steadily decreased):



The minimum possible value that can be chosen for L_m is obtained by **pinching the column**. A **pinch** occurs when there is **no driving force for mass transfer** (so $y = H_m x$) and the operating line hits the equilibrium line.

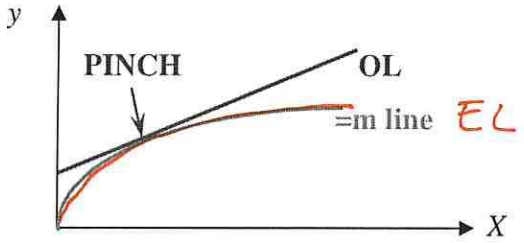
The pinch usually (but not always) happens at one end of the column: for absorption [scrubber] = pinch at the bottom; desorption [stripper] = pinch at the top.

E.g. for stripper:



The pinch will give the **minimum gradient allowed for the operating line, $(L_m/G_m)_{min}$.** However, the actual flow rate of L_m (or G_m for desorption case) is usually chosen so that the ratio of molar flow rates is between **$1.1 \rightarrow 1.4 \times (L_m/G_m)_{min}$.** This is due to economic considerations – if less than this is used, then the annual equipment costs increase at a higher rate than any savings in operating costs.

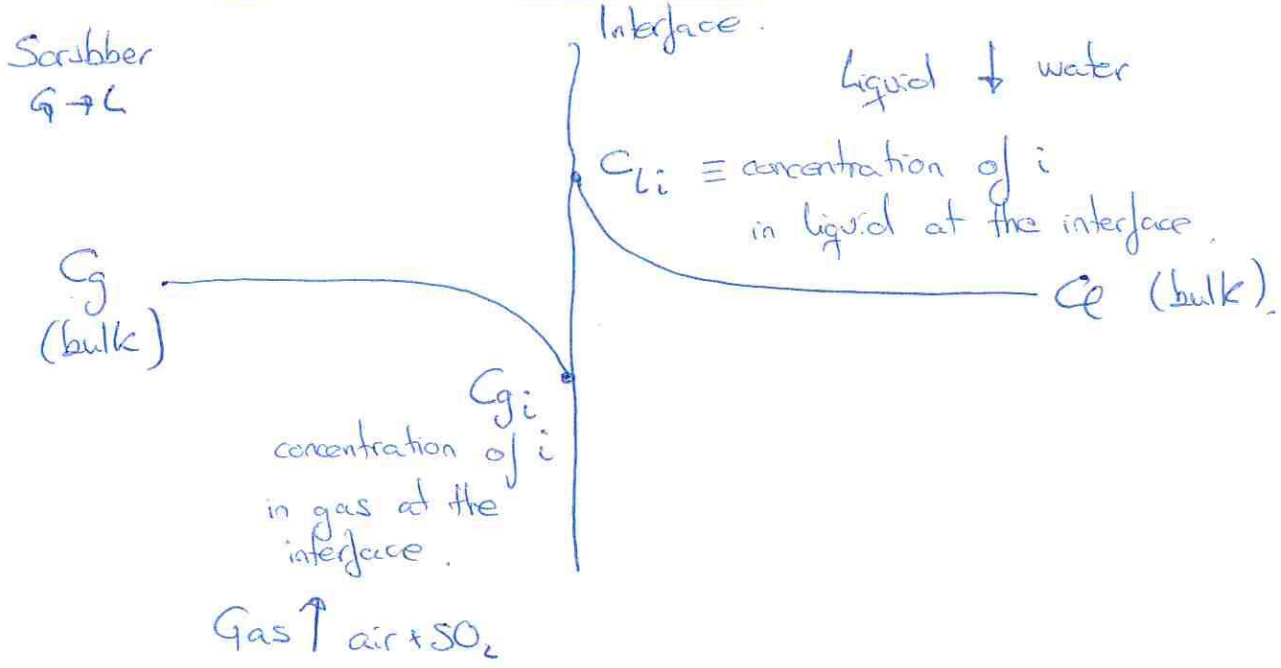
N.B. In practice, the **equilibrium line is <u>non-linear</u>**, *i.e.* Henry's Law is not applicable across the complete concentration range. In this case, the value of $(L_m/G_m)_{min}$ could be given when the operating line becomes tangential to the curve of the equilibrium line:



2.4. MASS TRANSFER

In most cases, the bulk liquid phase and gas phase will <u>not be in equilibrium</u>, and we have to consider the **rates of mass transfer** between them.

Consider the **interface** between the gas and liquid phases in an absorber, with solute moving



Experiments show that the rates of chemical dissolution of A (SO_2 here) into the liquid are much larger than the rates of diffusion. Therefore, we can assume that the **gas and liquid phases are at equilibrium at the interface itself**. Hence:

$C_{gi} = H_c \cdot C_{li}$ $H_c < 1$ because $C_{gi} < C_{li}$

In the **steady state**, the **flux *N*** [kmol m^{-2} s^{-1}] of A leaving the bulk gas phase equals that entering the liquid phase, *i.e.*

General equation: $N = K \cdot \Delta C \rightarrow$ driving force. (K: mass transfer coeff.)

Gas-side mass transfer

$$N = k_g \cdot (C_g - C_{gi})$$

$$\frac{N}{k_g} = (C_g - C_{gi}) \quad (1)$$

Liquid-side mass transfer

$$N = k_l (C_{li} - C_l)$$

$$\frac{N \cdot H_c}{k_l} = H_c \cdot C_{li} - H_c \cdot C_l$$

$$= C_{gi} - C_g^* \quad (2)$$

C_g^*: concentration in gas phase that would be in equilibrium with concentration in liquid phase.

(1) + (2) to eliminate unknown C_{gi}:

$$\frac{N}{k_g} = C_g - C_{gi} \quad (1)$$

$$+ \quad \frac{N \cdot H_c}{k_l} = C_{gi} - C_g^* \quad (2)$$

$$N\left(\frac{1}{k_g} + \frac{H_c}{k_l}\right) = C_g - C_g^*$$

$C_g - C_g^*$: driving force.

$\frac{1}{k_g} + \frac{H_c}{k_l}$: resistance.

$N = K_g \cdot (C_g - C_g^*)$

K_g: overall gas-side mass transfer coefficient $\left(\frac{m}{s}\right)$

$\frac{1}{K_g}$ = overall gas-side resistance

where

$$\frac{1}{K_g} = \frac{1}{k_g} + \frac{H_c}{k_l}$$

In this example, $C_g > C_g^*$, otherwise mass transfer would be in the other direction, *i.e.* from liquid to gas.

Alternatively, consider the overall situation from the liquid side:

Gas-side

$$N = k_g (C_g - C_{gi})$$

Divide by H_c

$$\frac{N}{H_c \cdot k_g} = \frac{C_g}{H_c} - \frac{C_{gi}}{H_c}$$

$$C_l^* - C_{li} \quad (1)$$

Liquid side.

$$N = k_L (C_{li} - C_l)$$

$$\frac{N}{k_L} = (C_{li} - C_l) \quad (2)$$

(1) + (2)

$$N\left(\frac{1}{H_c \cdot k_g} + \frac{1}{k_L}\right) = C_e^* - C_l$$

$$\frac{1}{K_L}$$

$$N = K_L \cdot (C_e^* - C_e)$$

Overall liquid-side mass transfer coefficient:

where $$\boxed{\frac{1}{K_l} = \frac{1}{k_l} + \frac{1}{H_c k_g} = \frac{1}{H_c K_g}}$$

Some typical values

Values of H_c at 20°C in water: $H_c \simeq 1$ $H_c \to 0$

N_2	O_2	CO_2	SO_2	NH_3
(insoluble)				(soluble)
60	30	1.05	9×10^{-3}	4.5×10^{-4}

For a typical system, $k_g \sim 0.04$ m s^{-1} and $k_l \sim 2 \times 10^{-4}$ m s^{-1} (the values do not depend on nature of gas too much).

Consider three gases:

i) CO_2 – **not very soluble**

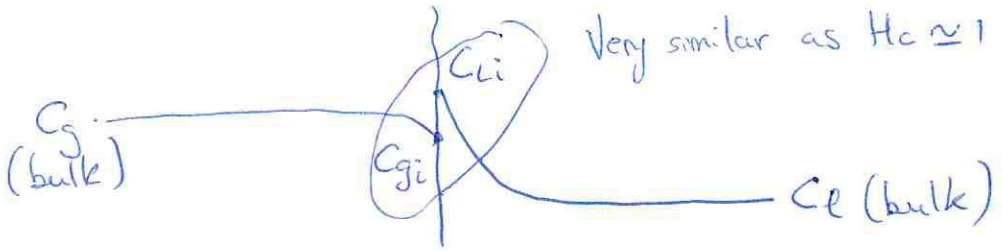
$$\frac{1}{K_l} = \frac{1}{k_l} + \frac{1}{H_c k_g} = \frac{1}{2\times10^{-4}} + \frac{1}{1.05\times0.04}$$

$$\Rightarrow \quad K_l = 1.99 \times 10^{-4} \text{ m s}^{-1} \approx k_l$$

$$K_g = \frac{K_l}{H_c} = 1.90 \times 10^{-4} \text{ m s}^{-1} \ll k_g$$

$K_l \approx k_l$, $K_g \ll k_g$ } negligible resistance in gas phase.

Therefore, the **resistance to mass transfer is on the liquid side**, *i.e.* $C_g \approx C_{gi}$:



ii) NH_3 – **very soluble**

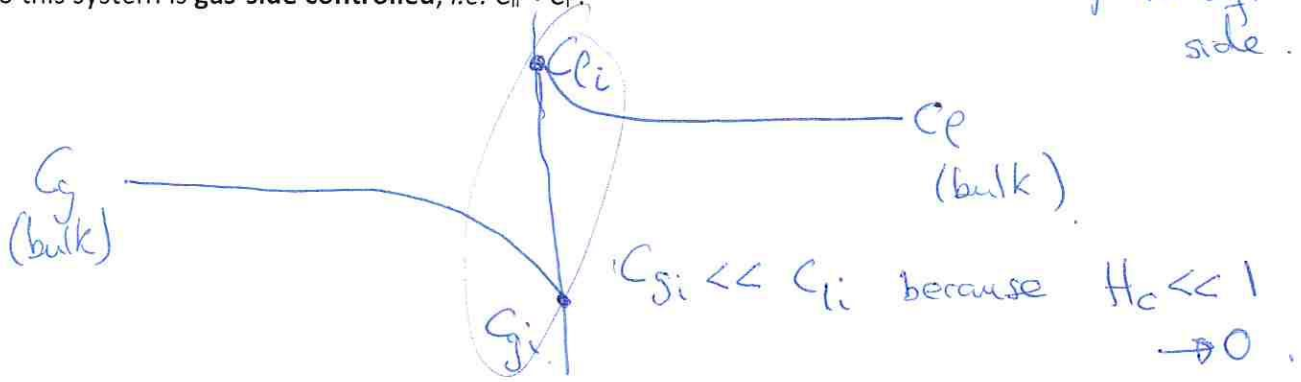
$$\frac{1}{K_l} = \frac{1}{k_l} + \frac{1}{H_c k_g} = \frac{1}{2\times10^{-4}} + \frac{1}{4.5\times10^{-4}\times0.04}$$

$$\Rightarrow \quad K_l = 1.65 \times 10^{-5} \text{ m s}^{-1} \ll k_l$$

$$K_g = \frac{K_l}{H_c} = 0.037 \text{ m s}^{-1} \approx k_g$$

$K_l \ll k_l$, $K_g \simeq k_g$. } negligible resistance in liq. side.

So this system is **gas-side controlled**, *i.e.* $C_{li} \approx C_l$:



iii) SO_2 – intermediate solubility

$$K_l = 1.3 \times 10^{-4}\ \text{m s}^{-1} \approx \tfrac{1}{2} k_l$$

$$K_g = 0.014\ \text{m s}^{-1} \approx \tfrac{1}{2} k_g$$

i.e. similar resistances on gas and liquid sides.

We can write the flux in terms of **mole fractions**:

G → L

$$N = k_g \cdot \rho_{mg} (y - y_i)$$

$$\frac{N}{k_g \cdot \rho_{mg}} = (y - y_i) \quad -(1)$$

$$N = k_L \cdot \rho_{mL} (x_i - x)$$

$$N = k_L \cdot \rho_{mL} \left(\frac{y_i}{H_m} - \frac{y^*}{H_m} \right)$$

$$\frac{N \cdot H_m}{k_L \cdot \rho_{mL}} = (y_i - y^*) \quad -(2)$$

(1) + (2)

$$N \left(\frac{1}{k_g \cdot \rho_{mg}} + \frac{H_m}{k_L \cdot \rho_{mL}} \right) = (y - y^*) \quad -(3)$$

p17 $$N = \bar{K}_g \cdot (C_g - C_g^*) = \bar{K}_g \cdot \rho_{mg} (y - y^*) \quad -4$$

$$\frac{N}{\bar{K}_g \cdot \rho_{mg}} = (y - y^*)$$

$$\frac{1}{\bar{K}_g \cdot \rho_{mg}} = \frac{1}{k_g \cdot \rho_{mg}} + \frac{H_m \cdot \rho_{mg}}{k_L \cdot \rho_{mL}}$$

Eq. liquid side:

$$\frac{1}{\bar{K}_L} = \frac{1}{k_L} + \frac{\rho_{mL}}{\rho_{mg} \cdot H_m \cdot k_g}$$

$$\bar{K}_L = \frac{\rho_{mg} \cdot H_m}{\rho_{mL}} \cdot \bar{K}_g$$

H_c

$$\bar{K}_L = H_c \cdot \bar{K}_g$$

P1 - P2

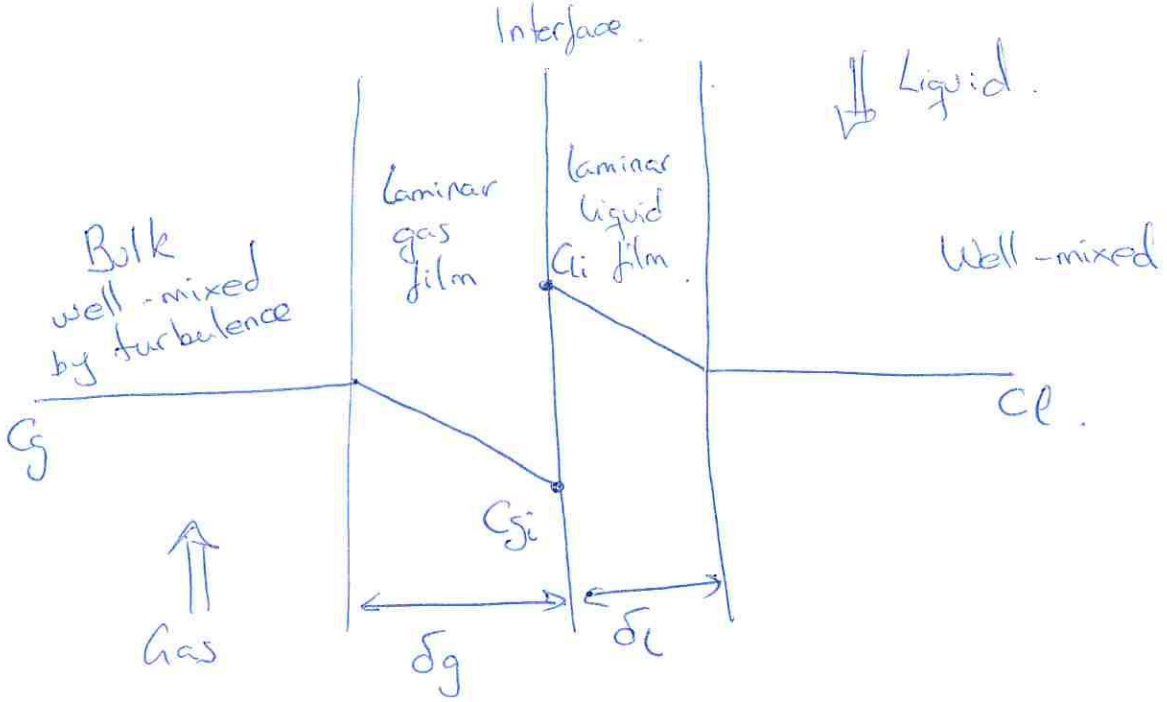
2.4.1. Whitman two-film model

We would like to predict K_g **and** K_l theoretically. The simplest approach is to extend the '**film model**' of mass transfer in a turbulent system.

Recall from your Heat and Mass transfer lectures:

$$\tfrac{1}{2} C_f = St' Sc \qquad \text{for '**film model**'}$$

This time we have two fluids present, and hence we need **two films**:



We assume that the concentrations vary **linearly** over the films.

Fick's law $\quad N = \frac{D}{\delta} \cdot \Delta C$

$D\ (m^2/s) \quad - \quad D_g = k_g \cdot \delta_g$

$D_L = k_L \cdot \delta_L$

From the film model: $\quad \tfrac{1}{2} C_f = St' Sc$

But $\quad Sh = Sc\ Re\ St' \quad$ (*cf.* $Nu = Pr\ Re\ St$ for heat)

Therefore $\quad Sh/Re = \tfrac{1}{2} C_f \quad$ But $C_f = f(Re)$

$Sh = \frac{k \cdot L}{D} \qquad Sh = 1/\delta$

So δ_g **and** δ_l **are functions of flow rates and are <u>independent of the component being absorbed</u>.**

We can measure K_g and K_l for <u>certain</u> gases using a wetted wall column.

$$Q_m = K_g A_s (\Delta C_g)_{lm} \qquad Q_m = K_l A_s (\Delta C_l)_{lm}$$

cf. **Heat:** $Q = UA(\Delta T)_{lm}$

$$(\Delta C_g)_{lm} = \frac{(C_{g,B} - C_{g,B}^*) - (C_{g,T} - C_{g,T}^*)}{\ln\left(\frac{C_{g,B} - C_{g,B}^*}{C_{g,T} - C_{g,T}^*}\right)}$$

<u>Case 1: H_c **is large** (insoluble gas),</u>

$$K_L \simeq k_L$$

So we can measure K_l for O_2 absorbing in water, say, which gives us an approximation for k_l.

Hence $\delta_l = \left(\frac{\mathcal{D}_l}{k_l}\right)_{O_2}$ [Need value for $(D_{O2})_l$]

For another gas A (the one we are interested in finding out about), $\delta_l = \left(\frac{\mathcal{D}_l}{k_l}\right)_A$

The δ **values are the same** (they are independent of the component being absorbed).

$$k_{L,A} = \left(\frac{D_A}{D_{O_2}}\right) \cdot k_{L,O_2}$$

<u>Case 2: H_c **is very small** (soluble gas, *e.g.* NH_3 absorbing in dilute H_2SO_4)</u>

$$K_G \simeq k_g$$

Therefore for A, $k_{g,A} = \left(\frac{\mathcal{D}_A}{\mathcal{D}_{NH_3}}\right)_g k_{g,NH_3}$ [Need value for $(D_{NH3})_g$]

Thus by knowing the diffusivities of A in gas and liquid, the $k_{l,A}$ and $k_{g,A}$ values can be estimated; then by knowing the H_c value, the $K_{l,A}$ **and** $K_{g,A}$ **values** can be predicted by:

$$\frac{1}{K_{l,A}} = \frac{1}{k_{l,A}} + \frac{1}{H_c k_{g,A}} \qquad \frac{1}{K_{g,A}} = \frac{1}{k_{g,A}} + \frac{H_c}{k_{l,A}}$$

2.5 SIZING OF ABSORPTION COLUMNS

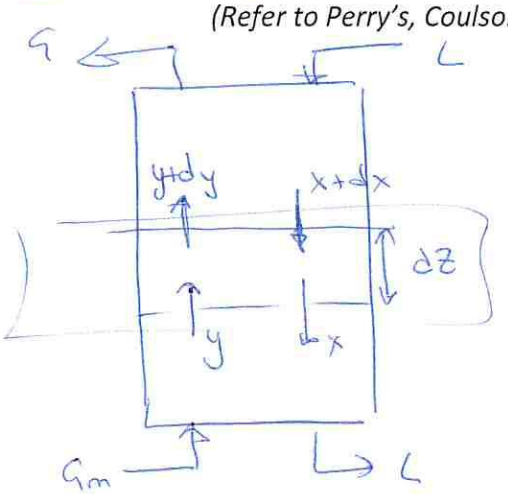
2.5.1 Transfer units

Introduce the **packing surface area per unit volume of packing, a [$m^2\ m^{-3}$]**. This is the wetted interfacial area in the packing available for mass transfer.

Some typical values: Random packing, $a \sim 100\ m^2\ m^{-3}$

Structured packing, a up to $500\ m^2\ m^{-3}$

(Refer to Perry's, Coulson & Richardson for more values)



Consider **overall gas-side transfer** over the height element dz :

G → L

In + Transferred = Out

$$G_m \cdot y - Q_m = G_m (y + dy)$$

$$Q_m = N \cdot a \cdot A \cdot V = A \cdot dz$$

$$G_m \cdot y - \underbrace{\rho_m \cdot K_g (y - y^*)}_{N} \cdot a \cdot \underbrace{A \cdot dz}_{V} = G_m (y + dy)$$

$$\int_0^z dz = -\int_{y_B}^{y_t} \frac{G_m \cdot dy}{\rho_{mg} \cdot K_g \cdot a (y - y^*) \cdot A}$$

$$z = \frac{G_m}{\rho_{mg} K_g a \cdot A} \cdot \int_{y_t}^{y_B} \frac{dy}{(y - y^*)}$$

This equation holds **irrespective of the equilibrium relationship**, and we can write:

$$z = \mathbf{HTU_{og}} \times \mathbf{NTU_{og}}$$

Height of a Transfer unit (overall gas-side)

Number of transfer unit (overall gas-side)

This analysis is an attempt to discretise the column (mathematically).

HTU_{og} is related to the **fluid mechanics** and to the **diffusional properties** obtained from correlations involving gas and liquid volumetric flow rates, *a*, diffusion coefficients, *etc.* It lies in the range 0.3 to 1.5 m.

NTU_{og} is related only to the **difficulty of the separation** and does not change with gas and liquid flow rate, choice of packing, *etc.* Generally, $NTU_{og} \neq$ equivalent number of plates.

Similarly, if we consider the situation from the **overall liquid-side**, applying:

G → L In + Absorbed = Out.

$$L_m(x+dx) + \underbrace{\rho_{mL} \cdot K_L \cdot (x^* - x)}_{N} \cdot a \cdot \underbrace{A \cdot dz}_{dV} = L_m \cdot x$$

$$L_m \cdot dx = -\rho_{mL} \cdot K_L \cdot (x^* - x) \cdot a \, A \cdot dz$$

$$\int_0^z dz = z = \frac{L_m}{\rho_{mL} \cdot K_L \cdot a \cdot A} \cdot \int_{x_T}^{x_B} \frac{dx}{(x^* - x)}$$

$$z = HTU_{ol} \times NTU_{ol}$$

$HTU_{og} \neq HTU_{ol}$ $NTU_{og} \neq NTU_{ol}$

$HTU_{og} \times NTU_{og} = HTU_{ol} \times NTU_{ol} = z$

Also note that:

$$NTU_{og} = \int_T^B \frac{dy}{y - y^*} = \int_T^B \frac{dp}{p - p^*} = \int_T^B \frac{dC_g}{C_g - C_g^*}$$

For a **straight equilibrium line**, *i.e.* H_m = constant,

$$K_L = \frac{H_m \cdot \rho_{mg}}{\rho_{mL}} \cdot K_g$$

$$HTU_{og} = \frac{G_m}{\rho_{mg} \cdot K_{og} \cdot a \cdot A} = \frac{G_m \cdot H_m \cdot \rho_{mg}}{\rho_{mg} \cdot \rho_{mL} \cdot K_L \cdot a \cdot A}$$

$$= \frac{G_m \cdot H_m}{L_m} \cdot \frac{L_m}{\rho_{mL} \cdot K_L \cdot a \cdot A}$$

$$\Rightarrow \quad HTU_{og} = \frac{G_m H_m}{L_m} HTU_{ol}$$

$\frac{G_m H_m}{L_m}$: A

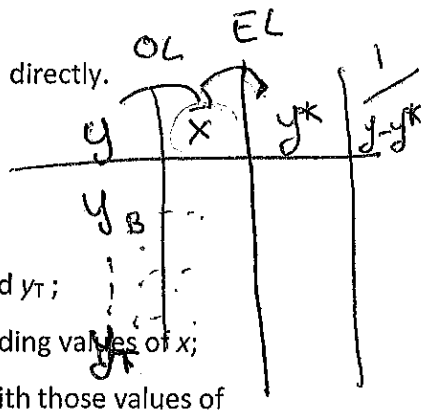
$\frac{L_m}{\rho_{mL} \cdot K_L \cdot a \cdot A}$: HTU_{ol}.

2.5.2. CALCULATION OF NTUs

Evaluation of NTU_{og}

$$NTU_{og} = \int_T^B \frac{dy}{y - y^*}$$

Both *y* and *y** vary up the column, so the integration cannot be performed directly.



Case 1: Curved operating and equilibrium lines:

The general method is:

i) Choose 5 or more values of *y*, including and lying between y_B and y_T;

ii) For each value of *y*, use the **operating line** to find the corresponding values of *x*;

iii) Use the **equilibrium line** to find the values of *y** in equilibrium with those values of *x*;

iv) Plot the graph of $\frac{1}{y - y^*}$ *vs.* *y* and find the area between y_B and y_T. (Or can use trapezium or Simpson's rule.)

Case 2: Straight operating and equilibrium lines:

In this case, the integration can be done analytically:

From **operating line:**

$$y = y_T + \frac{L_m}{G_m}(x - x_T)$$

Using $y^* = H_m x$ replace the *x* and x_T :

$$y = y_T + \frac{L_m}{G_m \cdot H_m}(y^* - y_T^*)$$ (A)

Rearranging to get *y** as subject:

Subtracting both sides of Eqn (A) from parameter *y* :

$$y^* = y_T^* + \frac{G_m \cdot H_m}{L_m}(y - y_T)$$

$$y - y^* = y - y_T^* - \frac{G_m \cdot H_m}{L_m}(y - y_T)$$

$$y - y^* = y\left(1 - \frac{G_m \cdot H_m}{L_m}\right) - y_T^* + \frac{G_m \cdot H_m}{L_m} y_T$$

$$y - y^* = y \cdot \alpha + \beta$$ — constant across the column.

So $$NTU_{OG} = \int_T^B \frac{dy}{y - y^*} = \int_T^B \frac{dy}{\alpha y + \beta} = \frac{1}{\alpha}\left[\ln(\alpha y + \beta)\right]_T^B$$

$$\Rightarrow \quad \boxed{NTU_{og} = \frac{1}{1 - \frac{G_m H_m}{L_m}} \ln\left(\frac{y_B - y_B^*}{y_T - y_T^*}\right)}$$ **For straight OL & EL**

Total amount of solute transferred in column:

$$Q_m = G_m (y_B - y_T) = L_m (x_B - x_T) \qquad y^* = H_m \cdot x$$

$$= \frac{L_m}{H_m} (y_B^* - y_T^*)$$

Therefore $$y_B - y_T = \frac{Q_m}{G_m}$$

and $$y_B^* - y_T^* = \frac{Q_m H_m}{L_m}$$

Subtracting both sides of these two eqns from each other:

$$(y_B - y_T) - (y_B^* - y_T^*) = Q_m \left(\frac{1}{G_m} - \frac{H_m}{L_m}\right)$$

$$(y_B - y_B^*) - (y_T - y_T^*) =$$

Multiplying through by G_m and rearranging:

Δy_B Δy_T

$$\frac{G_m (\Delta y_B - \Delta y_T)}{Q_m} = \left(1 - \frac{G_m \cdot H_m}{L_m}\right)$$

$$NTU_{OG} = \frac{Q_m}{G_m (\Delta y_B - \Delta y_T)} \cdot \ln\left(\frac{\Delta y_B}{\Delta y_T}\right) \rightarrow \frac{1}{(\Delta y)_{lm}}$$

$$z = \frac{G_m}{\rho_{mg} K_g a A} \frac{Q_m}{G_m} \frac{1}{(\Delta y)_{lm}} \cdot NTU_{OG}$$

But $z = HTU_{og} \times NTU_{og}$ $\Rightarrow$

$(HTU)_{OG}$

Solve for Q_m.

and so $$\boxed{Q_m = \rho_{mg} K_g a A z (\Delta y)_{lm}} \qquad (3.3)$$

N.B. Result only valid for **straight operating and equilibrium lines.** Q3 and Q4

Also note similarity to heat exchanger analysis.

Aside We could have developed expressions of the form:

$$z = \frac{G_m}{\rho_{mg} k_g a A} \int_{y_T}^{y_B} \frac{dy}{y - y_i} = \frac{L_m}{\rho_{ml} k_l a A} \int_{x_T}^{x_B} \frac{dx}{x_i - x}$$

i.e. $z = HTU_g \times NTU_g = HTU_l \times NTU_l$ **no 'overall' subscript**

$$HTU_{og} = \frac{G_m}{\rho_{mg} K_g a A} = \frac{G_m}{\rho_{mg} a A}\left(\frac{1}{k_g} + \frac{\rho_{mg} H_m}{\rho_{ml} k_l}\right) = \frac{G_m}{\rho_{mg} k_g a A} + \frac{G_m H_m}{L_m}\frac{L_m}{\rho_{ml} k_l a A}$$

$$\Rightarrow \quad HTU_{og} = HTU_g + \frac{G_m H_m}{L_m} HTU_l$$

Similarly $$HTU_{ol} = HTU_l + \frac{L_m}{G_m H_m} HTU_g$$

2.5.3 HEIGHT OF AN EQUIVALENT THEORETICAL PLATE (HETPs)

This is another method of calculating the height of a column. First use the methods in the Equilibrium Staged Processes course (CEBT 1B: Separations I) to find the **number of theoretical plates *n*** needed for the separation. Then estimate the height of the tower by:

$$\boxed{z = n \times HETP}$$

where HETP is the height of packing such that the streams leaving that section of the packing appear to be in equilibrium (*i.e.* $y = H_m x = y^*$).

This is just a numerical relationship between the streams, and true equilibrium is not being achieved.

This method was originally developed for distillation in packed columns, and involves **much empiricism**. Some typical HETP values for absorption (see *Coulson and Richardson Vol. 2*):

- Random packing 1.5 – 1.8 m
- Structured packing 0.1 – 0.3 m

Typical plate columns have tray spacings of about 0.5 m, and so are often shorter than random-packed columns.

RELATIONSHIPT BETWEEN HETP AND PACKED COLUMN HEIGHT

Case 1: Straight, but not parallel operating and equilibrium lines

From last year, the number of stages n required for gas absorption is given by:

$$n = \frac{\ln\left(\frac{\Delta y_B}{\Delta y_T}\right)}{\ln\left(\frac{y_T - y_B}{y_T^* - y_B^*}\right)} \qquad \text{----- (3.4)}$$

[derive from Kremser-Souders-Brown Equation, Data Book p.32]

If the actual height of the packed column is z, then $\text{HETP} = \frac{z}{n}$ ----- (3.5)

But from Equation (3.3), $z = \frac{Q_m}{\rho_{mg} K_g a A (\Delta y)_{lm}}$ ----- (3.6)

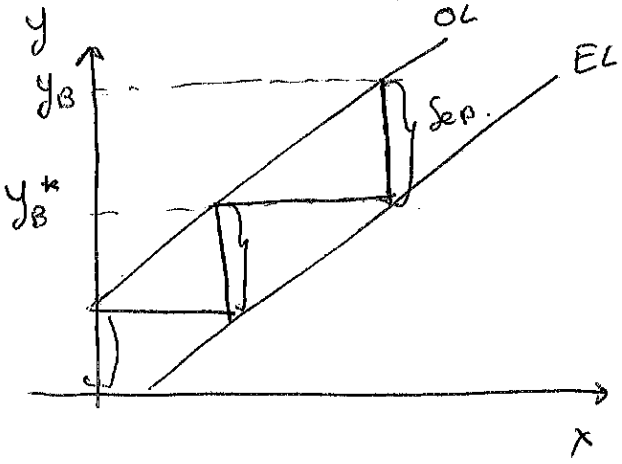
Equations (3.4 to 3.6) can be combined, along with the **overall operating line**, to give:

$$\text{HETP} = \frac{G_m}{\rho_{mg} K_g a A} \frac{\ln\left(\frac{L_m}{G_m H_m}\right)}{\left(1 - \frac{G_m H_m}{L_m}\right)}$$ — Q5

Case 2: Straight and parallel operating and equilibrium lines

$$\frac{L_m}{G_m} = H_m \quad \Rightarrow \quad \frac{L_m}{G_m H_m} = 1$$

and the value of *n* is **indeterminate**. In this case, *n* is merely the overall change in concentration divided by the driving force, which is constant.



Same separation in each stage.

$$n = \frac{y_B - y_T}{y_B - y_B^*}$$ — total separation / separation in each stage.

$$NTU_{og} = n$$

$$\text{NTU}_{og} = \int_T^B \frac{dy}{y - y^*} = \int_T^B \frac{dy}{y_B - y_B^*} = \frac{y_B - y_T}{y_B - y_B^*} = n$$

for this case only

and since $z = \text{HTU}_{og} \times \text{NTU}_{og}$, then from Equation (6) (**HETP = *z*/*n***) :

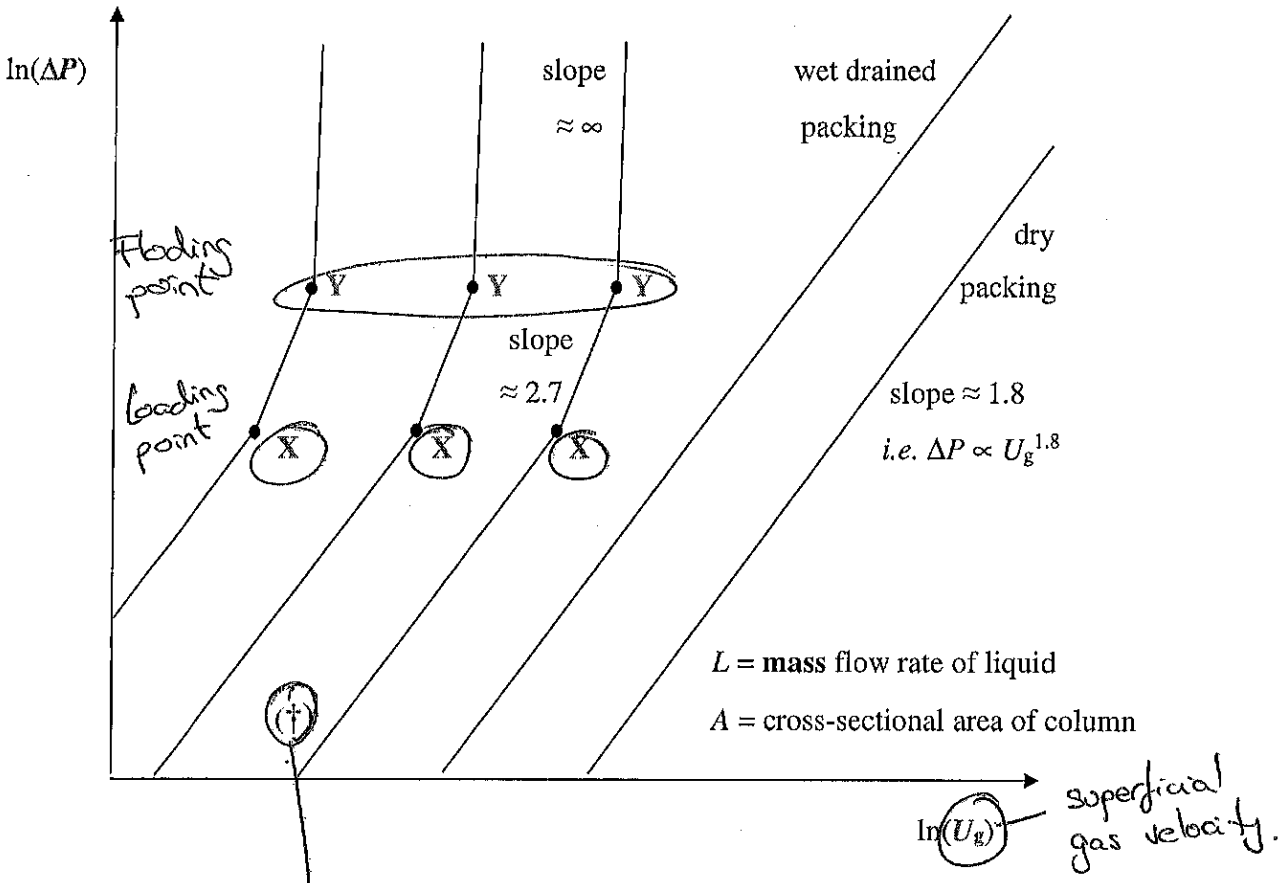
$$\text{HETP} = \frac{G_m}{\rho_{mg} K_g a A} = \text{HTU}_{og}$$

for this case only

2.6 OPERATING FEATURES

Selection of packed column diameter

For a given gas and liquid flow rate, the **diameter** of the column will affect the flow rates per unit cross-sectional area of the bed. The counter-current flow of gas and liquid results in **pressure drops** ΔP over the column. The following plot applies for a wide variety of packings.



For a given **superficial gas velocity** U_g, as L/A increases, the **pressure drop** ΔP increases since the liquid hold-up in the column reduces the space for gas flow.

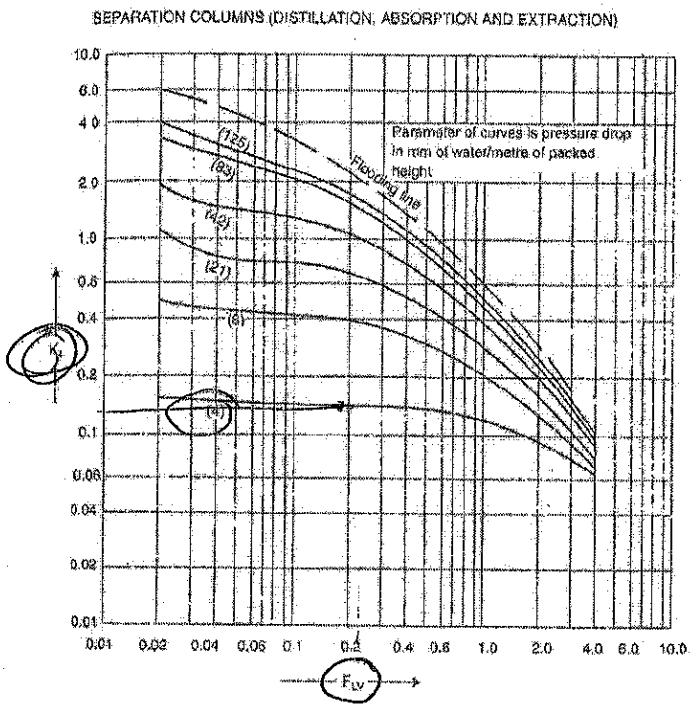
Looking at a typical curve **(†)**, at low U_g the packing acts like a series of orifices. Increasing L/A merely increases the thickness of the liquid film and hence decreases the space for gas flow. A point is reached where the slope of the curve increases with U_g, known as the **loading point (X)**. More liquid now becomes trapped in the packing as U_g increases, and consequently the slope of the above plot increases.

As U_g is increased further, the **gas moving upwards impedes the down-flowing liquid.** Eventually a point is reached when the liquid will not run downwards and the **column fills completely with liquid,** and so becomes inoperable. This is known as **flooding (Y),** and occurs when $\Delta P \rightarrow \rho g z$. ΔP is very high because gas cannot pass up through the column. The flooding point is more well-defined than the loading point.

So the gas and liquid velocities must be chosen to be less than that which causes flooding. It is usual to operate with ***U_g* about 50-60 % of its value for flooding,** so that the operation is at, or just below, the onset of loading. This allows for reproducible results, and is also the best compromise in terms of fixed and operating costs.

Pressure drops and flooding points are indicated from charts of the following form.

Image taken from Coulson & Richardson Vol. 6:



Where $$K_4 = \frac{13.1\left(\frac{G}{A}\right)^2 F_p \left(\frac{\mu_l}{\rho_l}\right)^{0.1}}{\rho_g\left(\rho_l - \rho_g\right)}$$ and $$F_{LV} = \frac{L}{G}\sqrt{\frac{\rho_g}{\rho_l}}$$

G, L = **mass** flow rate of gas, liquid [kg s^{-1}] A = cross-sectional area of column [m^2]

ρ_g , ρ_l = **mass** density of gas, liquid [kg m^{-3}] μ_l = dynamic viscosity of liquid [Pa s]

F_p = packing factor, depending on size and type of packing [m^{-1}]

Recipe for finding column diameter

i) By knowing absolute values of G and L [kg s^{-1}] (can obtain from G_m and L_m values [kmol s^{-1}]), calculate $\boldsymbol{F_{LV}}$.

ii) Use graph to find $\boldsymbol{K_{4,flood}}$.

iii) Choose an appropriate value of ΔP per unit length of packing, and find the corresponding K_4 value.

iv) Check that **0.5 < $K_4/K_{4,flood}$ < 0.6**, *i.e.* near but not too close to flooding.

v) Use this value of K_4 to calculate $(G/A)_{op}$ from the equation above.

vi) The column diameter is then given by $\sqrt{\frac{4G}{\pi}\Big/(G/A)_{op}}$.

vii) Check that the column diameter is at least 10× individual packing diameter.

Q6

Note that these types of plot are not very accurate, and different ones give predictions that can vary by 25%.

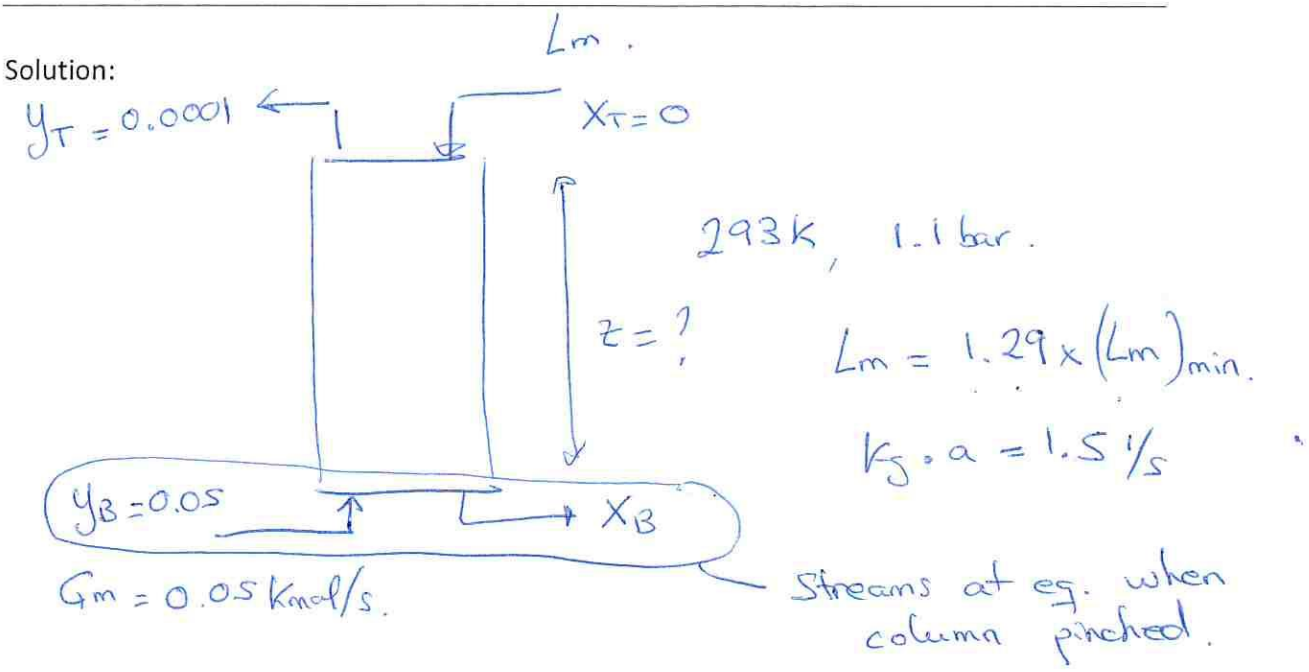
Worked example

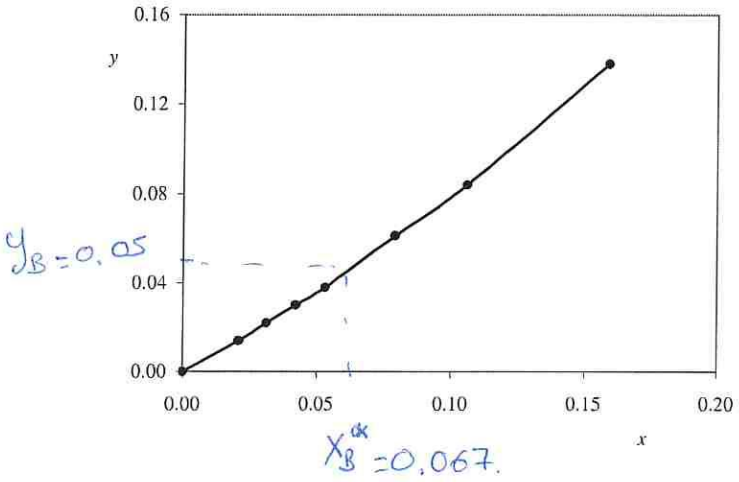
A process gas consists of 5 mol% ammonia-air mixture and is to be treated in a packed column by counter-current scrubbing with pure water at 293 K and 1.1 bara to reduce the ammonia content to 0.01 mol%. **Calculate the height of the absorber** if it operates at 1.29 $L_m)_{min}$, given that the gas flow rate is 0.05 kmol s^{-1}. Take $k_g a$ = 1.5 s^{-1}.

Equilibrium data for ammonia-water system: at 293K, 1.1bar.

x_{NH3}	0	0.021	0.031	0.042	0.053	0.079	0.106	0.159
p_{NH3} / kPa	0	1.6	2.4	3.3	4.2	6.7	9.3	15.2
$y_{NH3} = p_{NH3}/P_T$	0	0.014	0.022	0.030	0.038	0.061	0.084	0.138

Solution:

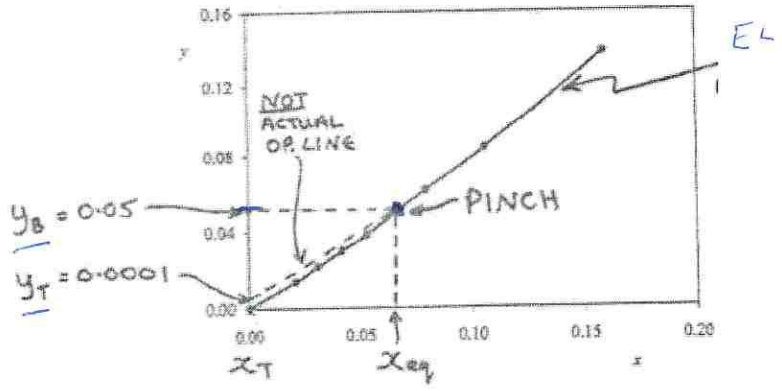




EL ≃ linear.

i. **Calculate $L_m)_{min}$ to find L_m**

A **pinch** situation will give $L_m)_{min}$ – the operating line will cross the equilibrium line, and equilibrium is established at the base of the column.



So the value of x_B at this pinch situation is x_{eq} = 0.067 (using linear interpolation).

Overall mole balance for pinch situation:

L_m ?

$$(L_m)_{min}\,(X_B^* - X_T) = G_m\,(y_B - y_T)$$

$$(L_m)_{min}\,(0.067 - 0) = 0.05 \times (0.05 - 0.0001)$$

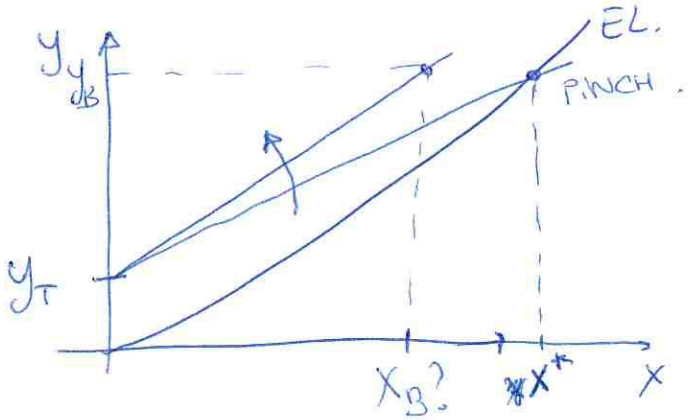
$$(L_m)_{min} = 0.037 \text{ kmol/s}.$$

$$L_m = 1.29 \times \text{ib} = 0.0480 \text{ kmol/s}.$$

ii. Calculate Q_m

***N.B.* x_B here now, not x_{eq}**

Overall balance: $\quad Q_m = G_m\,(y_B - y_T) = L_m\,(x_B - x_T)$



$$G_m\,(y_B - y_T) = L_m\,(X_B - X_T)$$

$$X_B = 0.0521$$

iii. **Find $(\Delta y)_{lm}$**

$$(\Delta y)_{lm} = \frac{(y_T - y_T^*) - (y_B - y_B^*)}{\ln\left(\frac{y_T - y_T^*}{y_B - y_B^*}\right)}$$

$x_T = 0 \quad y_T^* = 0$

$x_B = 0.0521 \quad y_B^* = 0.038$

$= 0.00249$

iv. **Calculate *z***

For **straight OL and EL**: $Q_m = \rho_{mg} K_g a A z (\Delta y)_{lm}$

Since ammonia is so soluble, $K_g \approx k_g$ (gas-side controlled).

Therefore $K_g a \approx k_g a = 1.5\ s^{-1}$

$$\rho_{mg} = \frac{P}{R.T} = \frac{1.1 \times 10^5}{8314.5 \times 293} = 0.0452\ kmol/m^3$$

$$A.z = \frac{Q_m}{\rho_m . K_g . a (\Delta y)_{lm}} = 14.8\ m^3$$

Proceed with analysis outlined in Section 2.6 in order to find suitable *A*. **Hence *z*.**

Alternative procedure *(replacing steps (iii) and (iv) above)*

Evaluate the NTU_{og} integral **numerically** (can use trapezium rule with varying step length).

This method allows for any **curvature in the equilibrium line.**

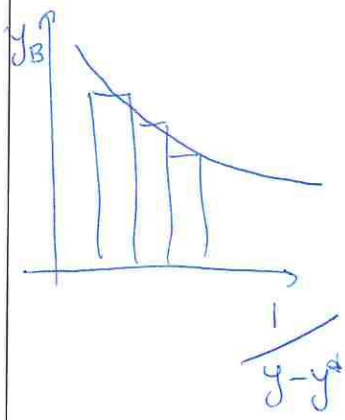
$$NTU_{og} = \int_T^B \frac{dy}{y - y^*} \qquad HTU_{og} = \frac{G_m}{\rho_{mg} K_g a A} \qquad z = HTU_{og} \times NTU_{og}$$

OL $L_m (x - x_T) = G_m (y - y_T)$

$$y = \left(\frac{L_m}{G_m}\right) x + y_T = 0.96x + 0.0001$$

OL EL

	y	*x* (from OL)	*y** (from EL)	1/(*y*-*y**)	Trapezium area
y_T	0.0001	0.0000	0	10000	
	0.0005	0.0004	0.0003	5000	3.00
	0.001	0.0009	0.0006	2500	1.88
	0.005	0.0051	0.0034	625	6.25
	0.01	0.0103	0.0069	323	2.37
	0.02	0.0207	0.0138	161	2.42
	0.03	0.0311	0.0221	127	1.44
	0.04	0.0416	0.0297	97	1.12
y_B	0.05	0.0520	0.0373	79	0.88
					Sum = 19.35



So trapezium rule gives $$\mathrm{NTU}_{og} = \int_T^B \frac{dy}{y - y^*} = 19.35$$

$$Z = HTU_{og} \times NTU_{og} = \frac{G_m}{P_{mg} K_g a A} \times 19.35$$

$$Z, \quad A = 14.3\ m^3.$$

Within 4% of the value of 14.8 m³ obtained assuming a straight equilibrium line